ITECH3102: Assessment Task 3:
Create a Static Visualization (20%)

Semester 2, 2026

Due date: Sunday, 23:59, Oct 4, 2026 - (Week 10)

1. Outline

Learning Outcomes Assessed K1, K2, K3

Purpose: To evaluate your ability to apply the basic theories and software you have learned in data visualization to design a simple visualization project.

Requirements: Design and create a simple visual project based on the given specification. This assessment task involves identifying related question, analysing tasks to be performed and selecting the most effective visual encoding to convert data values to graphical form.

Assessed by: Lecturer/tutor

Submission: Upload your work (report) to Moodle. The name of your work should be your name plus your ID.

2. Dataset

There are eight students who participate tests for nine subjects. Each of subjects has been marked from 0 to 100. The performance of the subjects that have been tested is shown in the following table:

Database	69	70	84	78	79	80	90	60
Java	64	74	58	83	55	72	67	59
C++	70	68	87	68	91	64	76	88
Physics	57	76	78	72	71	45	51	90
Mathematics	80	71	67	90	34	66	45	76
English	34	56	67	48	26	76	59	34
Chemistry	45	56	43	23	76	45	33	43
Networks	56	45	34	23	67	56	64	72
Security	96	30	56	46	75	54	34	45
Students	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**

3. Details of static visualization's requirement

For this assignment, you are responsible for designing a simple visual project where you have scores in nine subjects taken by eight students.

The requirements of this assignment are:

1) **Create datasets based on the data provided in the above table**
2) **Identify your questions**
   You should ask at least 3 questions from different aspects of data shown in this table. Your static visualization should effectively communicate different aspects of data and answer the questions you asked. To identify your questions, you may tell a story of the data and ask questions from the point of view of different angles such as teacher to know something about the subjects. For example, are there some subjects where students are doing particularly badly or well compared to the entire portfolio of subjects?
   Do NOT ask all your questions to be about correlations or min or max values, etc.
3) **Design your visualization**
   - Task abstraction
     Based on different aspects of data and the questions you asked, what is task abstraction to be made?
   - Data abstraction
     Based on different aspects of data and the questions you asked, what is data abstraction to be made? For example, do you need or is how the data set derived from data in the table in order to answer specific question?
   - Data types
4) **Convert data values to graphical form**
   You may select the most effective visual encoding to convert data values to graphical form. For example, what kinds of channels do you select to visualize your data? What kinds of marks such as color, area do you use in charts?
5) **Analyse your data visualisation**
   For example, why do you want visualise the data? How does your suggested visual encoding approaches to support effectively and efficiently answering the questions you asked above?

6) Write a visual report

Your written visual report should list at least 3 questions and answers, which includes a screen shot showing the visualization you used to answer each question, with one page per question, related screen shot and narrative. Each question should be answered with a different visualization.

4. Visualization Software

To create the visualization for this task, you will use Tableau or D3.js only. This assessment task will not be marked if other software is used in this task.

5. Assessment criteria for the assignment

Marking Criteria	Marks
Task and data extraction • You should show ability to extract data from the dataset table shown above. • Data is saved in one of proper formats. There are no questionable data entries.	**2**
Identify your questions • You should demonstrate ability to create questions that are very interesting. • Topic is of very interesting and shows creative thought in topic	**4**
Content of visualization • You should demonstrate ability to build of content of graphs that express knowledge, tells a story or answers questions in a compelling way • Your visualization reveals unknown information to audience.	**5**
Convert data values to graphical form You should demonstrate ability to create visualisation where type of visualisation methods chosen for the data (chart, marks, and channel) **support and enhance** the story or knowledge being conveyed.	**5**
Overall design structure and display • You should demonstrate ability to create visualisation whose design choices are clean, legible and • effectively help audience to get insight from data.	**4**
Total Marks	**/20**

6. Submission details

This is an individual assessment task. To submit your assignment, you must prepare a report including

- your visualization,
- related description,
- analysis,
- submit it to Moodle.

7. Late penalty

A 5% penalty will be applied to the total mark for each late day after the due date (including weekends and public holidays). Marking will cease after 7 days and the assessment will be scored at zero. If an assessment task is submitted beyond 7 days, it receives zero.

Feedback

Feedback will be given via Moodle. Marks are in fdlGrades and Moodle, and individual feedback considering each assessment criterion is provided via Moodle.

Academic misconduct

To submit your assessment task, you must indicate that you have read and understood, and comply with, the Federation University Australia Academic Integrity and Student Plagiarism policies and procedures (http://policy.federation.edu.au/learning_and_teaching/compliance/academic_integrity/ch02.php).

You must also agree that your work has not been outsourced and is entirely your own except where work quoted is duly acknowledged. Additionally, you must agree that your work has not been submitted for assessment in any other course or program.